Towards Vessel Arrival Time Prediction Through a Deep Neural Network Cluster

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Abstract. The prediction of accurate vessel arrival times is essential and challenging at the same time to plan vessel arrivals with sufficient accuracy, coordinate berthing manoeuvres and monitor ship traffic efficiently. This paper investigates a new approach by using clusters consisting of deep artificial neural networks (DNNC). For this purpose, the considered coverage area of the Weser river was divided into geospatial domains. An also developed linear regression model (LNNC) served as a reference model, which was generated analogously to the machine learning approach on the clusters. The estimated time of arrival prediction was evaluated at a distance of 50 km between the estuary of the Weser river into the North Sea and the target industrial port. It could be shown that the mean deviation from the actual travel time at a distance of 50 km is −19.80 min for the DNNC and 67.90 min for the LNNC. At a distance of 33 km from the industrial port, the mean deviation of the DNNC decreases to 2.85 min and for the LNNC to 54.40 min. Furthermore, it has been observed that the shorter the distance to the destination port, the more accurate the predictions become.

Keywords: vessel arrival time forecast · speed forecasting · machine learning · data-driven modelling

1 Introduction

Maritime shipping is one of the fastest-growing transport sectors worldwide. Up to 90% of the total trade volume per weight is transported by ship [1]. To be able to plan berthing processes and loading and unloading operations in a more energy-efficient and thus more sustainable and optimised way, precise traffic monitoring is necessary [2]. Research has already examined how waiting times of seagoing vessels lying at anchor affect environmental emissions. Among other things, proposals are derived that, through accurate arrival forecasts, seagoing

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vessels can reduce their speed early to reach the port and thus berth just in time. For this, an accurate arrival forecast is imperative. By reducing the speed of the ships, they reach the port of destination later but can therefore avoid unnecessary waiting times and reduce greenhouse gas emissions [3]. Shipping currently generates 1,076 million tonnes in 2018 of CO_2 equivalents per year and this figure is rising [4]. Franzkeit et al. (2020) conclude that reducing waiting times can not only reduce costs but also CO_2 emissions [5].

Accurate estimated time of arrival (ETA) forecasts of ships are essential for suitable berth allocation or personnel and infrastructure management. As part of a research project that serves as the underlying use case and the basis for the data presented in this paper, a current investigation is being conducted on how, through the implementation of low-threshold interfaces, such as REST-API, a networked port logbook can serve to make diverse information and data accessible to various stakeholders in the port sector. The method used in this paper to calculate vessel arrival times, which will continue to be integrated into this port logbook, represents a significant planning factor that can enable the efficient use of port infrastructure for various users, such as pilots and terminal operators.

In the context of this publication and the investigations carried out, an industrial port within the Weser, located approximately 50 river kilometres from the North Sea, serves as the application case. To achieve an optimised allocation of berths for ships, reliable forecasts regarding the arrival times of ships prove to be of paramount importance. These forecasts form the basis for deriving potential speed parameters, which can be used to fine-tune the allocation of berths [6]. This would not only lead to increased efficiency of the port infrastructure but also to a reduction in emissions of ship exhaust gases. However, this initial time estimated by the port authorities is often too imprecise for the use case. The port monitoring system used by authorities is continuously supplemented by staff with information from the ships and berth assignments, and lock times.

Frequently, the expected arrival times of vessels are, on the one hand, estimated based on the experience of port staff and, on the other hand, transmitted via radio communication at specific passing points from the captain to the port staff. Logging these estimated arrival times in a port logbook poses challenges, and their accuracy might be compromised. These passing points were selected as part of the evaluation of this work to evaluating the arrival forecasts depending on the corresponding distance to the industrial port. So our ETA predictions are calculated at these fixed passing points within the Weser river and the North Sea. The Weser was chosen as an exemplary region here, as the results generated in this paper are discussed in direct communication with port staff in this region and can be evaluated with the passing points stored in the port system.

The aim of this work was to determine how precise arrival times can be predicted using a deep neural network cluster. These results have been compared with a linear also cluster-based reference model. Ships approaching the port from a northerly direction do not have to pass through river locks but can sail directly to the destination port, provided that the tides, the draught of the ships and the ship traffic management allow this. For this reason, this route was chosen for our models. The predictions in this study are calculated using only dynamic and static ship information and without environmental data such as weather and tidal. To access data for the ML models to be trained continuously, the Automatic Identification System (AIS) has been accessed. These data contain dynamic and static information for each recorded ship. It is necessary to obtain the data from the AIS at an early stage and store it in data management systems to have a sufficient database for model generation. In this study, we could show that we could predict a more precise ETA with an artificial deep neural network cluster than with a linear regression model as reference.

2 Related Works

A notable amount of research has already appeared that has dealt with predicting ship arrival times and trajectories in terms of content. Often these sources refer to specific application areas such as the investigation of waiting time or effects on environmental emissions or to concrete geographical regions for which the predictions of ship movements or possible arrival times are considered. In the course of the research, the following works, in particular, were discovered, which are explained below and which, moreover, date back only a few years. Noman et al. (2021) showed that vessel arrival times on inland waterways could be predicted using a data-based approach and a gated recurrent unit neural network. On a large amount of AIS data, different models for ETA prediction of vessels on natural and artificial waterways have been developed [7]. Abebe et al. (2020) used AIS data and weather information to predict the speed of ships over ground (SOG) with machine learning (ML) methods. For this purpose, they observed 76 vessels over one year and trained and validated the models [8]. Abualhaol et al. (2018) attempted to predict the average vessel service time using a Long short-term memory (LSTM) based on AIS data. The port of Singapore was considered a port of interest in this work [9]. Hexeberg et al. (2017) have developed trajectory predictions of autonomous surface vessels (ASV) based on historical AIS data. Depending on the manoeuvrability of the ASV, the next position and associated time are determined in a time horizon of a couple of minutes up to 15 min using a data-based approach [10]. In their work, Brebisson et al. (2015) investigated how taxi destinations can be predicted using multi-layer perceptrons and bidirectional recurrent neural network architectures. Sequences of GPS (Global Positioning System) data and various other meta-information were used for this purpose. Compared to many other conventional approaches, an automated approach using neural networks was used here [11]. To the authors' knowledge, approaches for predicting vessel arrival times using data-based machine learning methods in natural river waterways are insufficiently explored. This paper describes a novel approach to predict ETAs from vessels in the river basin to an industrial port by dividing the coverage area into specific geospatial domains from which the clusters are formed.

3 Methodology and Material

This chapter explains the available and used data and steps for data preprocessing and provision for modelling to predict estimated arrival times. Furthermore, the developed models and their results are described, compared and discussed.

3.1 Data Acquisition

It is assumed that there is a causal relationship between the ship parameters to be identified, which enables an approximation of the arrival time via a speed forecast and the distance to the destination port. As a first step, therefore, the complete acquisition of all relevant information is necessary. As seen in Table 1 the following data and information from the AIS for the prediction of ship-specific speed over ground were identified during the process acquisition.

Table 1. Selected feature and label parameters in the AIS dataset

Feature: Course over Ground, Length, Breadth, Shiptype and Position
Label: Speed over Ground (SOG)

Automatic Identification System

The Automatic Identification System permanently records and transmits the ship's movements and other meta-information via sensors and transmitters on board the vessel. Depending on the current speed over ground, this data is transmitted at intervals of up to three minutes via two alternating channels in the Very High Frequency (VHF) maritime radio range. For the ETA forecast model developed in this paper, purely terrestrial AIS data sources have been used and obtained from a third-party provider. Satellite-AIS data was not available but was not required as the use case was close to the corresponding coastal and river region. The data sent from the AIS can be divided into dynamic and static information. Static vessel information includes parameters that cannot be changed, such as length, breadth, name, Maritime Mobile Service Identifier Number (MMSI) and type of vessel. These are only changed very rarely, for example, in the case of the sale of a vessel. The dynamic AIS information contains information on the current SOG, course over ground (COG), draught, and other [12]. In addition, the AIS data also contains manually entered information such as the current port of destination and a possible arrival time at that port. This information will not be considered in detail in this article, as it can be highly error-prone, for example, because it is outdated, too inaccurate or intentionally entered incorrectly by humans into the system [13]. For the training and testing of the ML models, 1,137 vessels have been identified and used in the period between December 2020 and June 2021. The ships placed in the coverage area have been assigned to the respective geospatial domain depending on their position. Abebe et al. (2020) were able to show that only a selection of the available parameters have a significant influence on the speed of a ship [8]. The presented work here has investigated how precisely the speeds of ships can be predicted if further environmental parameters such as weather and tidal data are omitted and whether a sufficiently accurate prediction is sufficient for the use case described in Sect. 1.

3.2 Data Preprocessing

After the data has been collected and archived, the ship information contained in the data records is preprocessed and erroneous data is filtered out. With the help of the responsible port authority, the required geospatial domains in Fig. 1 have been selected in the course of the river Weser and the estuary to the North Sea. The available dataset was used to assign the ships located in these areas to the respective geospatial domain. For each geospatial domain, the following data preprocessing steps take place before a deep neural network is developed and trained for each of these domains:

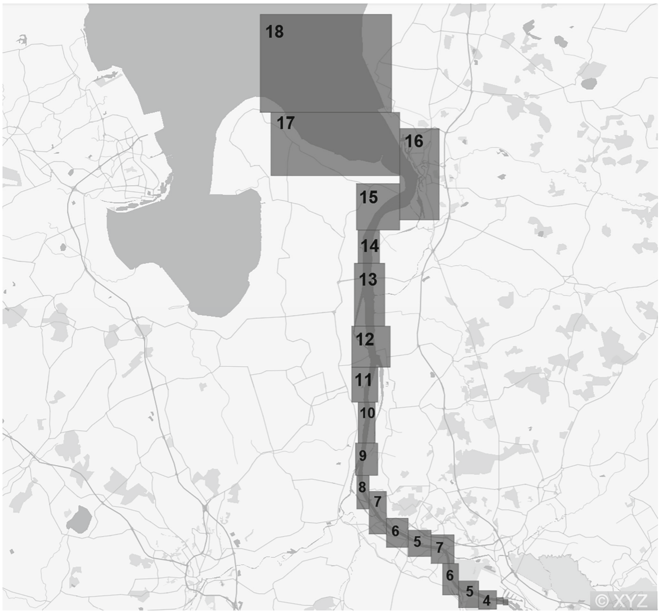

Fig. 1. Subdivided coverage in geospatial domains within the Weser river

1. In the data management system, the dynamic and static vessel informations are available separately for reasons of performance and storage space optimisation. In a first step of data preprocessing, these are again assigned to the respective ships.

2. The present dataset is separated from not required parameters (e.g. call sign of a vessel) so that only the parameters described in Table 1 remain in the dataset.
3. Non-moving or anchored vessels ($SOG < 2\ kn$) are filtered out of the dataset. Thus, on average, about 30 % of the ships remain in the dataset per geographic area.
4. As part of the data pre-processing, it was determined through expert interviews that fast-moving vessels ($SOG > 15\ kn$) would also be removed from the data set. This leads to a further reduction of the data volume to approx. 19.7 % with the mean value of the geographical area.
5. For the modelling of speed forecasts for seagoing and inland vessels, only container and cargo ships are required. Thus, based on the shiptype, ship classes that are not needed are removed from the dataset.
6. The dataset is further cleansed by filtering out ($COG = 511$) from erroneous course data.
7. The remaining and cleaned data is divided into a training and test dataset (80/20). Furthermore, the features are separated from the speed over ground label.
8. In the training data, random 20% of the data is taken to validate the generated neural network.
9. The input and output values from the dataset are standardised using the statistical parameters of the training data to implement a uniform scaling of the data.

These data preprocessing steps are repeated for all geospatial domains. This preprocessed data is used to generate and train the deep-neural-network-cluster (DNNC). Furthermore, this preprocessed dataset has been used for the generation of the regression model.

4 Prediction Models

The aim is to use a machine learning model to predict ship speeds precisely enough to calculate the most accurate possible time of arrival of vessels along the route to the destination industrial port, with reference to the associated distances between the route points. Although the ships are already within the Weser, speed over ground leads to fluctuations, so it can be suspected that only an inadequate arrival forecast can be produced with a linear formalism. As part of the data preprocessing, the course of the river and the estuary of the Weser to the North Sea have been divided into geospatial domains of different sizes. For each of these regions, an deep neural network has been developed, trained and evaluated. Thus, our developed deep neural network cluster is created by aggregating the models trained on the geospatial spaces. To be able to compare the deep neural network cluster generated in (A), we implemented a linear neural network cluster (B) as a reference model. The different evaluated arrival forecasts are generated and evaluated along the passage points of seagoing vessels recorded by the authorities, as seen in Table 2.

A. Deep Neural Network Cluster

The deep neural network cluster comprises a deep multi-layer perceptron neural network (MLP) for each defined area, as shown in Fig. 1. We used a deep feed-forward neural network with a specific depth and number of neurons per layer. Each of these layers receives its input from the previous layer. The DNNC is trained and validated using the preprocessed datasets. In the present research work, we have decided to resort to a cluster of deep artificial neural networks because of the large amount of data, about 1 GB per day, and the complex underlying real-world process to be mapped. Formally, such a dataset D can be represented as follows, so that x is represented as the input and y as the output to the network:

$$D = (x^1, y^1), (x^2, y^2), ..., (x^T, y^T) \tag{1}$$

The outputs a_j^l of each neuron j in the hidden layers l are calculated by the following formula:

$$a_j^l = f\left(\sum_{i=1}^{n} w_{ij}^l \cdot a_j^{l-1} + b_j^l\right) \tag{2}$$

Here w_{ij}^l and b_j^l present the respective weights and biases in layer l [14]. In this paper, Rectified Linear Unit (ReLU) is used as the activation function f. The output of the deep neural network is represented by a single neuron to predict a scalar value (in our case speed over ground) depending on the input parameters. The lower the calculated model error in the validation dataset, the better the model's performance. For the training of the networks, different training scenarios with different neural network configurations and hyperparameters have been selected and tested. The network structures with the most accurate predictions were selected. A time-consuming and computationally intensive grid search for the identification of hyperparameters was not performed in this work. For the chosen neural network architecture of the DNNC, the Rectified Linear Unit acts as the activation function of each neuron. This piecewise linear function forwards the output of the neuron if the result is positive, otherwise the output of the neuron is set to zero. Advantages of using a rectifier function are the simple implementation and calculation by using $f(z) = max(0, z)$. Furthermore, by using ReLU, true zero values can be generated when negative inputs occur. This simplifies the model and accelerates model learning.

The Adam algorithm has been used as an optimiser for the deep neural network cluster. Adam is a stochastic gradient descent method. Kingma et al., (2014) describe this algorithm as much more computationally efficient, invariant to diagonal rescaling of gradients and has significantly lower memory requirements than other optimisers [15]. For the DNNC, a learning rate of 0.001 has been applied as a model hyperparameter. To reduce the risk of generalisation during training of the neural network cluster and thus the risk of overfitting, a dropout layer has been added to the neural network. The dropout layer ensures that output connections of the neural network are disconnected randomly during training. Srivastava et al. (2014) explain that this method can reduce co-apdating of the individual neural units [16]. For the DNNC, a dropout rate of 0.11 has been set as model hyperparameter. Thus, with a probability of 11%, the weights of a neuron in this specific hidden layer are set to zero. The model has been tested with different hyperparameters as part of the modelling. For the DNNC, the number and position of the dropout layers and the dropout rate turned out to be optimal for the ETA calculation in this work. With the help of artificial neural networks, an optimal generalisation performance is to be achieved by training in order to achieve a sufficiently accurate approximation on the target data. All artificial neural networks tend to overfit [17]. While the neural network appears to get better during training, the model validation error gets worse with data unknown to the network, as overfitting to the training data occurs. To monitor and prevent overfitting, the neural network is examined during training with validation data and the early stopping method is used to terminate the training prematurely at the point where potential overfitting begins [18]. For this work, the patience model hyperparameter for the DNNC was set to 8 epochs. The patience model hyperparameter is typically used in training neural networks with techniques like early stopping. Early stopping is a regularisation technique that prevents overfitting in machine learning models.

B. Linear Neural Network Cluster
The Linear-Neural-Network-Cluster (LNNC) serves as a reference regression model to evaluate the results of the DNNC generated in (A). Analogous to (A), a linear model has been generated for the LNNC in each geospatial domain. For this purpose, each model in the cluster consists of a single neuron with a linear activation function. Due to a constant validation error during the training, the training for each geospatial domain could be terminated after only a few training cycles. Both models have been implemented with Keras and Tensorflow backend in Python. The DNNC and LNNC were trained and validated in a containerised Docker environment in a Python distribution (version = 3.6.10).

4.1 Results

The deep neural network cluster and the corresponding regression model have been trained with the preprocessed data for each geospatial domain. During the training of both models, the model used the mean squared error as an evaluation criterion.

A. Results of the DNNC
For the DNNC, the training in each geospatial domain is terminated after about 30 epochs by the early stopping, as from this point, overfitting of the model begins. It can be observed that the training loss in the region near the river mouth to the North Sea is calculated up to 1.5 knots SOG higher than further south of the Weser.

B. Results of the LNNC
In each geospatial domain, it is noticeable when generating the linear regression model that the training and validation error remains constant at about five knots after only a few training cycles. Figure 2 and Table 2 also indicate that, in

contrast to the DNNC, the LNNC allows for less accurate predictions. Especially in the case of more significant deviations from the mean speed of the ships in the corresponding domain, the predictions by the linear model are less precise.

C. Calculation and Evaluation of Ship Arrival Times by the DNNC and LNNC
From the preprocessed dataset, journeys of seagoing vessels to the port at 50 km (first passing point and contact with the port authorities of the vessels) have been retrospectively generated using the preprocessed dataset. At each passing point, where the vessels report their position to the authorities, a model prediction of the vessel arrival time has been calculated by the remaining distance and the associated route points. In addition, the actual time remaining to the port has been determined for each of these journeys at the respective passing point. Table 2 summarises the results of the arrival predictions at the four passing points of the seagoing vessels. The respective domains are also indicated in Table 2 to classify the corresponding passing points. As already assumed in (B), it can also be seen in Table 2 that the linear regression model calculates significantly less accurate predictions for the arrival time. For example, the DNNC has a mean difference between predicted and actual arrival time of about −20 min at the passing point *Nordenham*, while the LNNC has a mean difference of about 68 min. Figure 2 visualises the results in Table 2 through a boxplot. It can be seen that both models can determine better predictions the closer the ship is to its destination.

Table 2. Evaluation of arrival time predictions by the DNNC and LNNC for retrospectively generated journeys of seagoing vessels to the industrial port

Passing point	Nordenham	Brake	Farge	Moorlosen
Distance [km]	50	33	16	4
Number of vessels	32	34	36	39
Geo Area (see Fig. 1)	15	12	7	2
DNNC SOG Validation MSE [kn]	4.1	2.5	1.3	2.7
Mean Actual Travel Time (ATT) [min]	171	114.8	69.6	23.8
DNNC Mean Estimated Travel Time (ETT) [min]	190.8	111.9	62.2	14.7
LNNC Mean Estimated Travel Time (ETT) [min]	103	60.4	33.6	7.9
DNNC Delta Mean ATT-ETT [min]	−19.8	2.8	7.3	9.1
LNNC Delta Mean ATT-ETT [min]	67.9	54.4	35.9	15.8

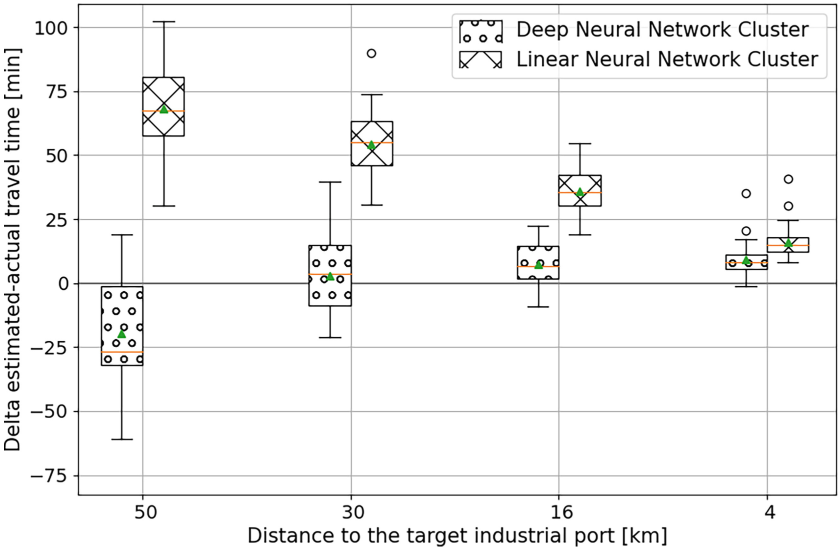

Fig. 2. Difference between predicted and real remaining travel time depending on the distance to the destination port at the four passing points

5 Conclusion and Discussion

In this work, we showed that using dynamic and static ship parameters, a sufficiently accurate prediction of ship arrival times, compared to a linear reference model and by manual estimation of port authorities, can be calculated by a deep neural network cluster. A comparison with a linear network cluster developed analogously has shown that far more precise predictions on the dataset used are possible through the DNNC. The models presented in this paper have not been post-trained. Thus, there is a possibility that seasonal factors may not be represented with sufficient accuracy by the models generated. For future work, it is possible to investigate how other influencing factors such as tidal range or weather information affect model performance. For both the DNNC and the LNNC, variations in velocity predictions can be seen in the preprocessed datasets. Nevertheless, the deep neural network cluster enables a sufficiently accurate ETA forecast to plan potential berthing times.

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